

Theta Asymptotics for High-Power Ballot Sums

OEIS A357825, A357871, and the growing-power array A357824

Exact oscillation envelopes, all-orders expansions, exponential multiset corrections, and inverse asymptotics

Abstract

Let $B(n, h)$ count nonnegative unit-step paths of length n ending at height h , and let $a_n = \sum_h B(n, h)^n$. The OEIS records the leading root growth of this sequence and the failure of a natural normalized ratio to converge. We derive its missing multiplicative structure: the ratio is a Jacobi theta function evaluated at a drifting parity-sensitive square-root phase, times $e^{-5/6}$. A uniform all-orders expansion expresses every subsequent coefficient as a Gaussian-weighted polynomial sum, equivalently a finite linear combination of theta derivatives. We prove the exact lower and upper limiting constants, identify the entire cluster interval and limiting phase law, and obtain oscillatory adjacent-ratio and Turán asymptotics. In particular, the sequence is eventually strictly log-convex.

The multiset counterpart $b_n = \sum_h \binom{B(n, h) + n - 1}{n}$ has the same algebraic asymptotic expansion after multiplication by $n!$, but differs by an explicitly determined first exponential sector of relative order $n^3 2^{-n}$. An exact rising-factorial identity supplies every fixed exponential sector. We also treat the array $\sum_h B(n, h)^k$: its transition from a continuous Gaussian approximation to a finite-width lattice theta sum occurs at $k \asymp n$. An exact two-endpoint estimate covers all larger powers and yields a second, near-square logistic transition at $k \asymp n^{3/2}$. Finally, we derive parity-sensitive inverse expansions and explain their precise relation to integer thresholds. The proofs are accompanied by independent exact integer checks, symbolic coefficient generation, and high-precision numerical experiments.

Status and scope. This is a self-contained mathematical research report, not a Lean formalization or an independently refereed paper. The asymptotic formulas below are proved here; computational checks support but do not replace those proofs. No claim of established publication priority is made. The prime-power supercongruence recorded for A357825 is *not* resolved in this report.

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1 The question, the contribution, and its provenance

1.1 What the OEIS already records

For a nonnegative integer n and an admissible height $h \in \{n, n-2, \dots, n-2\lfloor n/2 \rfloor\}$, put

$$B(n, h) = \binom{n}{(n-h)/2} - \binom{n}{(n-h)/2 - 1}, \quad (1.1)$$

where a binomial coefficient with lower index -1 is zero. These are the ballot numbers in [A0008315](#). Reflection at the first visit to height -1 gives the interpretation as nonnegative paths with steps $+1$ and -1 . Define

$$S_{n,k} = \sum_h B(n, h)^k, \quad a_n = S_{n,n}, \quad b_n = \sum_h \binom{B(n, h) + n - 1}{n}. \quad (1.2)$$

The array $S_{n,k}$ is [A357824](#), its diagonal a_n is [A357825](#), and b_n is [A357871](#) [4, 3, 1, 2]. The powers count ordered tuples sharing a common endpoint; the binomial coefficients count multisets sharing a common endpoint. The first values are

$$(a_n)_{n=0}^6 = (1, 1, 2, 9, 98, 4150, 562692), \quad (b_n)_{n=0}^6 = (1, 1, 2, 5, 21, 183, 3424).$$

For $n \geq 1$, set

$$K_n = \frac{2^{n+3/2}}{\sqrt{\pi e} n}, \quad \mathcal{A}_n = K_n^n = \frac{2^{n^2+3n/2}}{\pi^{n/2} e^{n/2} n^n}, \quad (1.3)$$

$$\mathcal{B}_n = \frac{e^{n/2} 2^{n^2+3n/2}}{\pi^{n/2} n^{2n+1/2}}. \quad (1.4)$$

Kotesovec's entries from November 2022 state $a_n^{1/n} \sim K_n$ and that $a_n/\mathcal{A}_n$ has no limit. The multiset entry gives the analogous root asymptotic and nonconvergence for $b_n/\mathcal{B}_n$ [1, 2]. Those observations are the starting point, not claims of this report's discovery. The natural next questions are: what replaces the missing constant, how large are the oscillations, and is there a uniform expansion beyond their leading shape?

1.2 What is established here

The main result is

$$\frac{a_n}{\mathcal{A}_n} = e^{-5/6} \left\{ \Theta(\alpha_n) + \frac{F_1(\alpha_n)}{\sqrt{n+1}} + \frac{F_2(\alpha_n)}{n+1} + \dots \right\}, \quad (1.5)$$

with a proved remainder after every fixed number of terms. Here

$$\alpha_n = \frac{\sqrt{n+1} - ((n+1) \bmod 2)}{2}, \quad \Theta(\alpha) = \sum_{j \in \mathbb{Z}} e^{-4(j-\alpha)^2}.$$

The coefficient functions are periodic and explicitly computable. The oscillations are therefore organized by a simple phase, not erratic noise. The main extensions are an exact limiting interval and distribution, eventual log-convexity, the $k \asymp n$ transition for the entire array, all fixed exponential sectors connecting the tuple and multiset counts, and an inverse expansion retaining the same phase.

Question or assertion	Status in this report
Root growth and absence of a limit	Already recorded by OEIS; proved and sharpened here.
Theta factor and every algebraic order	Explicit formulas with a uniform localization proof.
Multiset exponential corrections	Exact finite identity and every fixed-sector remainder.
Growing powers and inverse formulas	Proved in the parameter ranges stated below.
Global log-convexity at every index	Not claimed; only eventual strict log-convexity is proved.
Bala's prime-power supercongruence	Left open; no arithmetic conclusion is inferred from asymptotics.
Lean verification and publication priority	Neither is claimed.

1.3 Relation to the literature and ProveIt

Catalan-triangle moments and fixed powers are part of an established literature; Miana and Romero study such moments, closed expressions in several cases, and divisibility questions [7]. The present asymptotic regime instead lets the exponent grow with the row index. Discrete Laplace methods are also established; a recent systematic formulation is given by Helton, Hughes, and Schlosser [8]. The point here is not to rename that method as new, but to retain the lattice when its mesh does *not* become small relative to the peak width. Standard Stirling and theta identities are used with attribution [9, 10, 11, 12].

Repository inspection used ProveIt commit

50f93367b2859b2727c34c68cf87684c101768cc

and its combinatorics map [13]. No existing repository theorem is required for the proofs below. This article is a proposed additional research case study; it does not promote a repository computation into a formal theorem. A repository code search for A357825 and targeted public searches for the sequence identifiers and theta/high-power ballot asymptotics did not identify a matching all-orders formula in the material reviewed. That is a limited search result, not a proof of novelty. The source-status audit and bibliography are dated 1 October 2026.

1.4 Place in the ProveIt collection

[Write note, batch 73O1, 1 October 2026.] This report is filed in ProveIt's research-report collection, in the directory

SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/
oeis-sequence-asymptotics/a357825-theta-ballot-power-sums.

Its README there is the guide to the files.

Provenance. The report is built from one manuscript: manuscript 59 of batch 73 (cluster O1) of ProveIt's incoming-reports intake, the archive OEIS_Theta_Ballot_Sums.zip, which arrived in commit aa43cc555 and was placed by commit 9df4ba51a. Its pin is the commit 50f93367b quoted above (the delivered source audit calls it the "inspected tree commit"; it is a commit). The text is printed as delivered. The write changed only the label names, which

now carry the prefix `tbs:`, and added the notes marked *[Write note]*. Nothing was merged into it, and no other manuscript of the batch treats these sequences.

Status. The manuscript was prepared with ChatGPT. It is unrefereed, and nothing in it is formalized in Lean or any other proof assistant. The intake reran the shipped verifier on a copy (`code/verify.py -all`, exit code 0; the eight data files it writes equal the delivered ones, up to line endings). It also checked independently the first terms of a_n and b_n , the two envelope constants of Theorem 4.2 to 20 digits, and the expansion through F_2 against exact sums for $n = 100, 200, 400, 800$. It did not re-derive every proof. The OEIS values and attributions quoted in the article were not re-fetched by the intake; they are as the manuscript inspected them on 1 October 2026.

Shipped layout. The delivered `requirements.txt` is shipped as `data/requirements.txt`, and `build.sh` as `code/build.sh`; the paths printed in Section 9 are the delivered ones. The figure PDFs were regenerated by the intake with embedded TrueType instead of Type 3 fonts, from the same data; see the README. The delivered README, PDF and checksum list are not shipped.

Neighbouring material. The same mechanism, a continuous saddle whose lattice is kept when its mesh stays comparable to the peak width, producing a theta multiplier, appears in two packages of Provel's series-and-transseries collection, for different objects: the q -multinomial package `Theta_Resolved_Optimal_Truncation_q_Multinomial` and the chapter "Galois numbers and root-lattice theta sectors" of `Combinatorial_Transseries_Inverses`. They are cited for comparison only; no statement here depends on them, and none of theirs on this report. The report `hankel-determinants/catalan-and-ballot/ballot-polynomial-hankel-determinants` studies Hankel determinants of ballot moments, not growing powers. Bala's supercongruence (question Q9 of Section 10) remains open.

2 The shifted ballot identity and the main theorem

2.1 Why the shift by one matters

Put

$$m = n + 1, \quad r = h + 1, \quad \varepsilon_n = m \bmod 2 \in \{0, 1\}. \quad (2.1)$$

Then r ranges over the positive integers of parity m with $r \leq m$. A binomial simplification gives the exact identity

$$B(n, h) = \frac{r}{m} \binom{m}{(m-r)/2}. \quad (2.2)$$

Define the continuous positive interpolation, only as an analytic tool, by

$$\mathfrak{b}_m(r) = \frac{r}{m} \frac{\Gamma(m+1)}{\Gamma((m-r)/2+1)\Gamma((m+r)/2+1)}, \quad 0 < r \leq m, \quad (2.3)$$

and introduce

$$V_m = \frac{2^m}{m} \sqrt{\frac{2}{\pi e}} = K_n \frac{n}{m}. \quad (2.4)$$

The maximum of the leading continuous shape is at $r = \sqrt{m}$. Writing

$$\alpha_n = \frac{\sqrt{m} - \varepsilon_n}{2}, \quad u = r - \sqrt{m} = 2(j - \alpha_n), \quad (2.5)$$

turns the summation into a translated lattice of fixed spacing 2. The phase can be reduced modulo 1 whenever a periodic function is applied to it. Keeping $m = n + 1$, rather than replacing it prematurely by n , is necessary for the correct constant and correction terms.

Definition 2.1 (Theta sums and Gaussian moments). For $\tau > 0$ and $p \geq 0$, let

$$\Theta_\tau(\alpha) = \sum_{j \in \mathbb{Z}} e^{-4\tau(j-\alpha)^2}, \quad M_{p,\tau}(\alpha) = \sum_{j \in \mathbb{Z}} [2(j-\alpha)]^p e^{-4\tau(j-\alpha)^2}. \quad (2.6)$$

Write $\Theta = \Theta_1$ and $M_p = M_{p,1}$. These sums and all of their phase derivatives converge absolutely and locally uniformly in $\tau > 0$. They are smooth, 1-periodic functions of α .

Theorem 2.2 (All-orders diagonal expansion). *There are explicitly computable polynomials $R_j \in \mathbb{Q}[u]$, with $R_0 = 1$, $\deg R_j \leq 3j$, and $R_j(-u) = (-1)^j R_j(u)$, such that for every fixed $J \geq 0$,*

$$\frac{a_n}{\mathcal{A}_n} = e^{-5/6} \left\{ \sum_{j=0}^J m^{-j/2} F_j(\alpha_n) + O_J(m^{-(J+1)/2}) \right\}, \quad m = n+1, \quad (2.7)$$

where

$$F_j(\alpha) = \sum_{\ell \in \mathbb{Z}} e^{-u_\ell^2} R_j(u_\ell), \quad u_\ell = 2(\ell - \alpha). \quad (2.8)$$

The implied constant is independent of the phase. In particular $F_0 = \Theta$, and the same order holds as a relative error after division by the leading term.

The first two correction polynomials are

$$R_1(u) = \frac{u^3}{3} + \frac{2u}{3}, \quad (2.9)$$

$$R_2(u) = \frac{u^6}{18} - \frac{u^4}{36} + \frac{11u^2}{9} + \frac{13}{60}. \quad (2.10)$$

Consequently

$$F_1 = \frac{1}{3}M_3 + \frac{2}{3}M_1 = \frac{1}{192}\Theta''' + \frac{7}{24}\Theta', \quad (2.11)$$

$$F_2 = \frac{1}{18}M_6 - \frac{1}{36}M_4 + \frac{11}{9}M_2 + \frac{13}{60}\Theta. \quad (2.12)$$

All derivatives here are with respect to α . At integer and half-integer phases F_1 vanishes by symmetry. Away from those phases an $m^{-1/2}$ term is generally present; restricting an ansatz to integral powers of $1/n$ loses genuine information.

Remark 2.3 (Meaning of “all orders”). Equation (2.7) is a Poincaré asymptotic expansion: J is fixed while $n \rightarrow \infty$. No convergence of the infinite series, optimal truncation theorem, or resurgent description is asserted. Because its coefficients are periodic functions of a drifting phase, it is an *oscillatory* expansion, not an ordinary real logarithmic-exponential transseries with constant coefficients. The additional exponential sectors in Section 7 give a separate, precise beyond-all-orders statement.

3 Uniform localization and the proof of the expansion

3.1 A Stirling expansion with a moving argument

Define

$$f(x) = \log x - \frac{x^2 - 1}{2}, \quad x > 0. \quad (3.1)$$

It has a unique maximum 0 at $x = 1$, with $f''(1) = -2$. The logarithmic Stirling expansion for Γ gives the following uniform form.

Lemma 3.1 (Uniform local logarithm). *For every compact interval $I \subset (0, \infty)$ and every integer $L \geq 0$,*

$$\log \frac{b_m(\sqrt{m}x)}{V_m} = f(x) + \sum_{\ell=1}^L A_\ell(x) m^{-\ell} + O_{I,L}(m^{-L-1}), \quad x \in I, \quad (3.2)$$

for all sufficiently large m . The estimate may be differentiated any fixed number of times in x , after increasing the finite Stirling truncation used to establish it. Here

$$\begin{aligned} A_\ell(x) = & -\frac{x^{2\ell+2}}{(2\ell+2)(2\ell+1)} + \frac{x^{2\ell}}{2\ell} \\ & + \sum_{j=1}^{\lfloor (\ell+1)/2 \rfloor} \frac{B_{2j}}{2j(2j-1)} \left[\mathbf{1}_{\ell=2j-1} - 2^{2j} \binom{2\ell-2j}{2\ell-4j+2} x^{2\ell-4j+2} \right], \end{aligned} \quad (3.3)$$

where B_{2j} denotes a Bernoulli number, with $B_2 = 1/6$.

Proof. Set $z = x/\sqrt{m}$, and insert the logarithmic Stirling formula into (2.3). The exponential part of the binomial coefficient is controlled by

$$\frac{(1+z)\log(1+z) + (1-z)\log(1-z)}{2} = \sum_{q \geq 1} \frac{z^{2q}}{2q(2q-1)}.$$

Its $q = 1$ term gives $-x^2/2$, and its higher terms give the first term of (3.3). The square-root prefactor contributes $-\frac{1}{2} \log(1-z^2)$, producing the second term.

For a Bernoulli term with $p = 2j - 1$, subtraction of the two denominator corrections produces

$$\frac{B_{2j}}{2jp} m^{-p} [1 - 2^p \{(1-z)^{-p} + (1+z)^{-p}\}].$$

The even Taylor coefficient of z^{2k} in the braces is $2^{\binom{p+2k-1}{2k}}$. Setting $k = \ell - p$ yields the summation in (3.3). The leading prefactors, together with r/m , are exactly $V_m \exp f(x)$.

For x in a fixed compact interval, both denominator arguments are $m/2 + O(\sqrt{m})$, hence are bounded below by a positive multiple of m . Finite Stirling remainder bounds and finite Taylor remainders are therefore uniform. The corresponding digamma and polygamma expansions supply the same conclusion after any fixed number of differentiations; see [9, 10]. This proves the finite asymptotic statement, without requiring convergence of the displayed infinite Taylor or Stirling series outside their use as coefficient generators. $\square$

For example,

$$\begin{aligned} A_1(x) &= -\frac{x^4}{12} + \frac{x^2}{2} - \frac{1}{4}, \\ A_2(x) &= -\frac{x^6}{30} + \frac{x^4}{4} - \frac{x^2}{3}, \\ A_3(x) &= -\frac{x^8}{56} + \frac{x^6}{6} - \frac{x^4}{3} + \frac{1}{24}. \end{aligned} \quad (3.4)$$

In particular $A_1(1) = 1/6$ and $A'_1(1) = 2/3$.

3.2 A global Gaussian bound

A local Stirling calculation alone does not justify exchanging an asymptotic expansion with a sum whose range grows. The necessary tail control follows from global concavity.

Lemma 3.2 (Uniform Gaussian domination). *There are constants $C, c > 0$ such that, for all sufficiently large n and all $0 < r \leq m = n + 1$,*

$$n \log \frac{b_m(r)}{K_n} \leq C - c(r - \sqrt{m})^2. \quad (3.5)$$

In particular the normalized summands of a_n have a phase-uniform summable Gaussian envelope.

Proof. Write $\ell_m(r) = \log b_m(r)$. Direct differentiation gives

$$\ell_m''(r) = -\frac{1}{r^2} - \frac{1}{4} \left[\psi_1\left(\frac{m-r}{2} + 1\right) + \psi_1\left(\frac{m+r}{2} + 1\right) \right], \quad (3.6)$$

where ψ_1 is the trigamma function. For $z > 0$ its positive series satisfies

$$\psi_1(z) = \sum_{q=0}^{\infty} (z+q)^{-2} \geq \int_0^{\infty} (z+t)^{-2} dt = \frac{1}{z}.$$

The two arguments in (3.6) sum to $m+2$, so their reciprocals sum to at least $4/(m+2)$. Thus $\ell_m''(r) \leq -1/(m+2)$ everywhere on the stated interval.

Lemma 3.1, including its differentiated form, gives

$$\ell_m'(\sqrt{m}) = \frac{2}{3m^{3/2}} + O(m^{-5/2}), \quad n \log \frac{b_m(\sqrt{m})}{K_n} = O(1).$$

Taylor's theorem with the global upper bound on the second derivative now gives, with $u = r - \sqrt{m}$,

$$n \log \frac{b_m(r)}{K_n} \leq C + O(m^{-1/2})u - \frac{n}{2(m+2)}u^2.$$

Completing the square absorbs the linear term and proves (3.5). $\square$

For $|u| > m^{1/12}$, the total contribution is

$$O\left(\exp(-c'm^{1/6})\right),$$

uniformly in the lattice translate, by the Gaussian envelope. This is smaller than every fixed algebraic power of m^{-1} . It also controls every fixed polynomial moment of the tails.

3.3 The coefficient engine

Let $t = m^{-1/2}$ and introduce a formal series in t whose coefficients are polynomials in u :

$$\begin{aligned} \mathcal{L}(t, u) &= -1 + (t^{-2} - 1) \left[f(1+ut) + \sum_{\ell \geq 1} A_\ell (1+ut) t^{2\ell} \right] + \sum_{\ell \geq 1} \frac{t^{2\ell}}{\ell(\ell+1)} \\ &= -\frac{5}{6} - u^2 + \sum_{j \geq 1} D_j(u) t^j. \end{aligned} \quad (3.7)$$

Only finitely many terms are required to obtain any specified coefficient. The last sum appears because

$$(m-1)\log(1-1/m) = -1 + \sum_{\ell \geq 1} \frac{m^{-\ell}}{\ell(\ell+1)}. \quad (3.8)$$

Thus $\mathcal{L}$ generates precisely the expansion of $n \log(b_m(\sqrt{m}+u)/K_n)$. The first exponent polynomials are

$$\begin{aligned} D_1(u) &= \frac{u^3}{3} + \frac{2u}{3}, \\ D_2(u) &= -\frac{u^4}{4} + u^2 + \frac{13}{60}, \\ D_3(u) &= \frac{u^5}{5} - \frac{2u^3}{3} - \frac{8u}{15}, \\ D_4(u) &= -\frac{u^6}{6} + \frac{u^4}{6} + \frac{2u^2}{3} + \frac{59}{420}. \end{aligned} \quad (3.9)$$

Exponentiation gives the exact polynomial recurrence

$$R_0(u) = 1, \quad R_j(u) = \frac{1}{j} \sum_{i=1}^j i D_i(u) R_{j-i}(u) \quad (j \geq 1). \quad (3.10)$$

Indeed it is the coefficient recurrence for $\exp(\sum_{i \geq 1} D_i t^i) = \sum_{j \geq 0} R_j t^j$. Each D_i has parity i and degree at most $i+2$. Any product contributing to weight j therefore has degree at most $3j$ and parity j , proving the structural assertions of Theorem 2.2.

The constant $-5/6$ is worth isolating. Raising the peak's Stirling correction to the n th power contributes $+1/6$ to the logarithm. Converting V_m^n to K_n^n contributes -1 . Their sum is $-5/6$; omitting either effect gives the wrong oscillation envelope.

3.4 A summable remainder and completion of the proof

Fix J . On $|u| \leq m^{1/12}$, apply Lemma 3.1 with a finite truncation larger than the order needed for t^{J+1} , and apply Taylor's theorem to the finitely many functions at $x = 1+ut$. Since $|ut| \leq m^{-5/12}$, these Taylor expansions are uniform. The first nonconstant perturbation of the Gaussian exponent is bounded by $Ct(1+|u|^3)$, which is $o(1)$ on this window. Finite Taylor's theorem for the exponential consequently gives some integer d_J and constants $C_J, c_J > 0$ such that

$$\begin{aligned} & \left| \left(\frac{b_m(\sqrt{m}+u)}{K_n} \right)^n - e^{-5/6-u^2} \sum_{j=0}^J m^{-j/2} R_j(u) \right| \\ & \leq C_J m^{-(J+1)/2} (1+|u|)^{d_J} e^{-c_J u^2}. \end{aligned} \quad (3.11)$$

To see why increasing the Stirling truncation suffices, its remainder before multiplication by n is $O(m^{-L-1})$ uniformly for x near 1; after multiplication it is $O(m^{-L})$. Choosing $L \geq \lceil (J+1)/2 \rceil$ and retaining the required Taylor terms makes it no larger than the requested order, with any extra polynomial factors absorbed into $(1+|u|)^{d_J}$. The finitely many exponentiation remainders have the same form.

For every $p \geq 0$ and $c > 0$,

$$\sup_{\alpha \in [0,1]} \sum_{j \in \mathbb{Z}} (1 + |2(j-\alpha)|)^p e^{-c[2(j-\alpha)]^2} < \infty.$$

Summing (3.11) over the translated lattice therefore preserves its algebraic order. Lemma 3.2 controls the omitted actual summands outside the window. The polynomial-Gaussian approximants have tails of the same negligible kind. Extending the finite admissible lattice to all of $\mathbb{Z}$ adds only such tails, since both endpoints are a growing distance from $\sqrt{m}$. This proves (2.7). Finally, the nearest lattice point has $|u| \leq 1$, so $\Theta(\alpha) \geq e^{-1}$ uniformly; absolute and relative versions of the theorem are equivalent. $\square$

4 Theta geometry, exact envelopes, and phase distribution

4.1 Theta identities and extrema

Poisson summation of a Gaussian gives

$$\Theta_\tau(\alpha) = \frac{\sqrt{\pi}}{2\sqrt{\tau}} \sum_{q \in \mathbb{Z}} \exp\left(-\frac{\pi^2 q^2}{4\tau}\right) e^{2\pi i q \alpha}. \quad (4.1)$$

This also follows from the standard theta transformation [12]. Termwise differentiation of (2.6) gives

$$M'_{p,\tau} = -2pM_{p-1,\tau} + 4\tau M_{p+1,\tau}. \quad (4.2)$$

For $\tau = 1$, iteration expresses every F_j as a finite rational linear combination of derivatives of Θ . This is an alternative all-orders representation to the rapidly convergent moment sums.

Lemma 4.1 (Strict extrema). *For each $\tau > 0$, Θ_τ is strictly decreasing on $(0, 1/2)$. Its global maximum is at integer phases and its global minimum is at half-integer phases.*

Proof. Let $q = \exp(-\pi^2/(4\tau)) \in (0, 1)$. Apart from the positive constant in (4.1), the Jacobi product is

$$\prod_{j \geq 1} (1 - q^{2j})(1 + 2q^{2j-1} \cos(2\pi\alpha) + q^{4j-2}).$$

Every phase-dependent factor is positive. Its logarithmic derivative is strictly negative when $0 < \alpha < 1/2$. The derivative series converges locally uniformly there, so the logarithmic derivative of the product is negative. Periodicity and reflection symmetry finish the proof. The product identity is standard [11]. $\square$

Theorem 4.2 (Exact limiting interval). *The cluster set of $a_n/\mathcal{A}_n$ is precisely $[C_-, C_+]$, where*

$$C_- = e^{-5/6} \sum_{j \in \mathbb{Z}} e^{-(2j-1)^2} = 0.31986675953099572612\dots, \quad (4.3)$$

$$C_+ = e^{-5/6} \sum_{j \in \mathbb{Z}} e^{-4j^2} = 0.45051819401966197435\dots \quad (4.4)$$

In particular these are respectively the liminf and limsup, and no constant C satisfies $a_n \sim C\mathcal{A}_n$.

Proof. Theorem 2.2 reduces the claim to the range of Θ along the phases. Lemma 4.1 supplies universal limiting bounds. For $n = s^2 - 1$, the phase is an integer exactly, so this subsequence converges to C_+ . For $n = s^2$, the phase tends modulo one to $1/2$, so that subsequence converges to C_- . Proposition 4.3 below proves that every phase occurs as a subsequential limit. Continuity of Θ then fills the entire interval and excludes other cluster values. $\square$

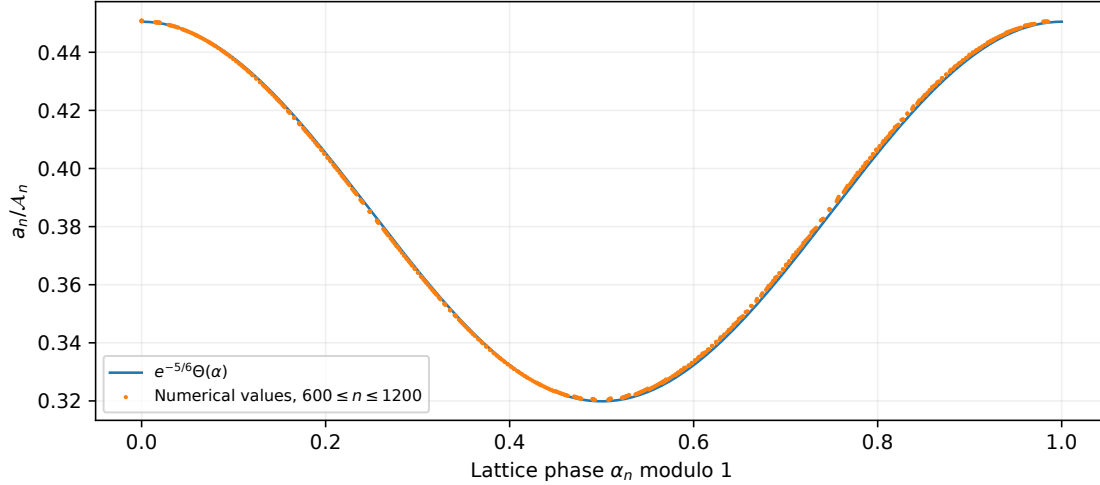


Figure 1: The apparent oscillation collapses onto a single theta curve when plotted against the phase. Dots are high-precision localized log-Gamma evaluations for $600 \leq n \leq 1200$, not interval-certified values. The discrepancy is explained by the $m^{-1/2}$ and higher terms.

4.2 An elementary equidistribution proof

Proposition 4.3 (Effective phase distribution). *For every interval $I \subset [0, 1]$,*

$$\#\{1 \leq n \leq N : \{\alpha_n\} \in I\} = N|I| + O(\sqrt{N}). \quad (4.5)$$

The implied constant can be chosen independently of I . The same assertion holds on either parity class, with its own number of terms in place of N .

Proof. Fix $\varepsilon \in \{0, 1\}$ and restrict $m = n + 1$ to that parity. For $I = [a, b]$, the condition is equivalent to membership in the union of intervals

$$(2j + \varepsilon + 2a)^2 \leq m < (2j + \varepsilon + 2b)^2,$$

apart from finitely many harmless initial endpoints. The number of integers of a specified parity in an interval of length L is $L/2 + O(1)$. The displayed interval has length

$$8j(b - a) + 4\varepsilon(b - a) + 4(b^2 - a^2).$$

There are $O(\sqrt{N})$ complete intervals up to $N + 1$. Summing their main terms gives $(b - a)$ times the number of admissible m up to $N + 1$, with error $O(\sqrt{N})$: the terms linear in j produce the quadratic main term, and all endpoint and linear-size corrections are of order $\sqrt{N}$. A final partial interval also has length $O(\sqrt{N})$ and obeys the same bound. All constants are uniform for $0 \leq a \leq b \leq 1$. Summing the two parity classes proves (4.5). $\square$

Corollary 4.4 (Limiting law and mean). *If an index is chosen uniformly from $1, \dots, N$, then $a_n/\mathcal{A}_n$ converges in distribution to $e^{-5/6}\Theta(U)$, where U is uniform on $[0, 1]$. Moreover*

$$\frac{1}{N} \sum_{n=1}^N \frac{a_n}{\mathcal{A}_n} = e^{-5/6} \frac{\sqrt{\pi}}{2} + O(N^{-1/2}) = 0.38515263413250459606 \dots + O(N^{-1/2}). \quad (4.6)$$

Proof. Continuity and equidistribution prove convergence in distribution. Integrating the discrepancy bound against the smooth function Θ gives an $O(N^{-1/2})$ error in its average. The average of the uniform $O(n^{-1/2})$ remainder in Theorem 2.2 has the same order. Finally

$$\int_0^1 \Theta(\alpha) d\alpha = \frac{1}{2} \int_{-\infty}^{\infty} e^{-u^2} du = \frac{\sqrt{\pi}}{2}.$$

□

The root asymptotic in the OEIS follows immediately: $a_n/\mathcal{A}_n$ is bounded above and bounded away from zero, so $a_n^{1/n}/K_n = (a_n/\mathcal{A}_n)^{1/n} \rightarrow 1$. The multiplicative oscillation survives in the ratio but disappears under the n th root.

5 Adjacent quotients, log-convexity, and recurrence obstructions

Put

$$\lambda = \log 2, \quad c = \frac{3}{2} \log 2 - \frac{1 + \log \pi}{2}, \quad g(x) = \lambda x^2 - x \log x + cx. \quad (5.1)$$

Then $g(n) = \log \mathcal{A}_n$. Consecutive phases satisfy

$$\alpha_{n\pm 1} = \alpha_n + \frac{1}{2} + O(n^{-1/2}) \pmod{1}. \quad (5.2)$$

Thus averaging away the phase before forming an adjacent quotient would lose an order-one factor.

Theorem 5.1 (Adjacent and Turán quotients). *As $n \rightarrow \infty$,*

$$\frac{a_{n+1}}{a_n} = \frac{2^{2n+5/2}}{\sqrt{\pi} e^{3/2} n} \frac{\Theta(\alpha_n + 1/2)}{\Theta(\alpha_n)} (1 + O(n^{-1/2})), \quad (5.3)$$

$$\frac{a_{n-1}a_{n+1}}{a_n^2} = 4 \left(\frac{\Theta(\alpha_n + 1/2)}{\Theta(\alpha_n)} \right)^2 + O(n^{-1/2}). \quad (5.4)$$

The second quotient has lower and upper limits

$$T_- = 4 \left(\frac{\Theta(1/2)}{\Theta(0)} \right)^2 = 2.01638539749423251125 \dots, \quad (5.5)$$

$$T_+ = 4 \left(\frac{\Theta(0)}{\Theta(1/2)} \right)^2 = 7.93499100910135656311 \dots \quad (5.6)$$

In particular a_n is strictly log-convex for all sufficiently large n .

Proof. Taylor expansion of g gives

$$e^{g(n+1)-g(n)} = \frac{2^{2n+5/2}}{\sqrt{\pi} e^{3/2} n} (1 + O(n^{-1})), \quad e^{g(n-1)+g(n+1)-2g(n)} = 4(1 + O(n^{-1})).$$

Use Theorem 2.2, (5.2), smoothness, and positivity of Θ . The ratio of two theta values is bounded by $\Theta(1/2)/\Theta(0)$ and its reciprocal. Both bounds are attained at phases 0 and $1/2$, which occur as subsequential limits.

The strict inequality $T_- > 1$ need not rest on floating-point decimals. For instance $\Theta(1/2) \geq 2/e > 2/3$, while

$$\Theta(0) = 1 + 2 \sum_{j \geq 1} e^{-4j^2} \leq 1 + \frac{2e^{-4}}{1 - e^{-12}} < \frac{11}{10}.$$

The last elementary bound follows already from $e > 8/3$. Consequently $T_- > 4(20/33)^2 > 1$, a positive gap from the log-convexity threshold. Equation (5.4) proves the final assertion. $\square$

A rapidly growing sequence need not automatically be log-convex: oscillatory multipliers can interfere with a second difference. Here the theta quotient is explicitly controlled, so the conclusion has a genuine proof rather than following from growth alone. The theorem does not specify the least index from which log-convexity holds.

Proposition 5.2 (No polynomial-coefficient linear recurrence). *Neither (a_n) nor (b_n) is P-recursive over $\mathbb{C}$. Equivalently, neither ordinary generating series is D-finite as a formal power series. Both ordinary generating series have radius of convergence zero.*

Proof. A sequence u_n satisfying a nonzero polynomial-coefficient recurrence $\sum_{j=0}^r p_j(n)u_{n+j} = 0$ can, for all large n , be solved for the term with largest nonzero index. Ratios of its polynomial coefficients are bounded in magnitude by Cn^d for some integer $d \geq 0$. If $M_N = \max(1, |u_0|, \dots, |u_N|)$, it follows that $M_{N+1} \leq C'N^d M_N$ after adjusting constants and ignoring a finite initial segment. Hence

$$|u_n| \leq C_0 C_1^n (n!)^d, \quad \log^+ |u_n| = O(n \log n).$$

A recurrence of order zero would force eventual vanishing and is also impossible here. Theorem 2.2 gives $\log a_n = \lambda n^2 + O(n \log n)$. Theorem 7.1 below gives the same quadratic leading logarithm for b_n . Both contradict the recurrence bound. The equivalence with formal D-finiteness follows by coefficient extraction from a linear differential equation and the converse conversion of a recurrence to such an equation. Finally $a_n^{1/n}$ and $b_n^{1/n}$ tend to infinity, so the radius assertion follows from the root test. $\square$

6 The growing-power array and two localization transitions

The scale of the exponent determines whether the lattice can be replaced by an integral. This section treats $S_{n,k}$ for real $k > 0$ analytically; integer k gives the OEIS array. The parameter $m = n + 1$ is always retained.

6.1 Fixed powers and the broad Gaussian regime

Proposition 6.1 (Fixed and subcritical powers). *For each fixed $k > 0$,*

$$S_{n,k} \sim V_m^k \frac{\sqrt{m}}{4} e^{k/2} \left(\frac{2}{k}\right)^{(k+1)/2} \Gamma\left(\frac{k+1}{2}\right). \quad (6.1)$$

If $k = k(n) \rightarrow \infty$ and $k = o(m)$, then

$$S_{n,k} \sim V_m^k \frac{\sqrt{\pi m/k}}{2}. \quad (6.2)$$

Proof. For fixed k , use $x = r/\sqrt{m}$. Its lattice mesh is $2/\sqrt{m}$. Lemma 3.1 gives pointwise convergence of $(b_m(\sqrt{m}x)/V_m)^k$ to $e^{kf(x)} = e^{k/2}x^k e^{-kx^2/2}$. The concavity proof gives a uniform integrable Gaussian bound in $x - 1$; compact-interval convergence and this tail bound justify the Riemann sum. Thus

$$S_{n,k}/V_m^k \sim \frac{\sqrt{m}}{2} \int_0^\infty e^{k/2} x^k e^{-kx^2/2} dx,$$

whose integral is (6.1).

For growing $k = o(m)$, use the finer local variable $v = (r - \sqrt{m})\sqrt{k/m}$. The mesh $2\sqrt{k/m}$ tends to zero, while

$$kf(1 + v/\sqrt{k}) \longrightarrow -v^2, \quad kA_1(1 + v/\sqrt{k})/m \longrightarrow 0$$

uniformly for bounded v . The same concavity argument gives a uniform Gaussian tail in v . The Riemann-sum limit is therefore $\frac{1}{2}\sqrt{m/k} \int_{\mathbb{R}} e^{-v^2} dv$, proving (6.2). $\square$

For orientation, the first two integral powers also have exact familiar identities:

$$S_{n,1} = \binom{n}{\lfloor n/2 \rfloor}, \quad S_{n,2} = \frac{1}{n+1} \binom{2n}{n}.$$

The first telescopes in (1.1). For the second, join a pair of nonnegative n -step paths with common height by following the first and then the reverse of the second. This is a bijection with Dyck paths of length $2n$. These columns are already identified by the OEIS [3]; they are consistency checks, not new enumeration claims.

6.2 The critical theta regime

Theorem 6.2 (Critical growing powers). *Let $\tau = k/m$ stay in a compact subinterval of $(0, \infty)$. Uniformly in τ and in the phase,*

$$S_{n,k} = V_m^k e^{\tau/6} [\Theta_\tau(\alpha_n) + O(m^{-1/2})]. \quad (6.3)$$

There is an expansion to every fixed order in $m^{-1/2}$, with coefficients that are polynomial-Gaussian sums and smooth in τ . Its first correction inside the brackets is

$$\frac{\tau}{\sqrt{m}} \sum_{j \in \mathbb{Z}} \left(\frac{u_j^3}{3} + \frac{2u_j}{3} \right) e^{-\tau u_j^2}, \quad u_j = 2(j - \alpha_n). \quad (6.4)$$

Proof. Replace the factor $n = m - 1$ in the local logarithmic analysis by $k = \tau m$. Lemma 3.1 gives

$$k \log \frac{b_m(\sqrt{m} + u)}{V_m} = \tau \left(\frac{1}{6} - u^2 \right) + \frac{\tau}{\sqrt{m}} \left(\frac{u^3}{3} + \frac{2u}{3} \right) + O(m^{-1}(1 + |u|)^4)$$

on bounded windows, uniformly for τ in the given compact interval. All higher coefficients follow by the same finite Taylor procedure. The global concavity bound, now multiplied by k , gives a summable Gaussian with constants uniform on that interval. The argument of (3.11) therefore applies without change. $\square$

The width in r is of order $\sqrt{m/k}$. It is large in (6.2), but of order one in (6.3). This explains the first localization transition, $k \asymp m$.

For a probabilistic formulation, choose an endpoint with probability $\mathbb{P}_{n,k}(h) = B(n, h)^k / S_{n,k}$. After writing $u = h + 1 - \sqrt{m}$, the critical law is approximated in total variation, with error $O(m^{-1/2})$, by the phase-shifted discrete Gaussian

$$\mathbb{P}\{u = 2(j - \alpha_n)\} = \frac{e^{-4\tau(j-\alpha_n)^2}}{\Theta_\tau(\alpha_n)}. \quad (6.5)$$

Here the comparison extends the admissible finite lattice to all integers; the added tail is negligible. The weighted remainder bound in the proof also yields

$$\mathbb{E}_{n,k}[h + 1 - \sqrt{m}] = \frac{\Theta'_\tau(\alpha_n)}{4\tau\Theta_\tau(\alpha_n)} + O(m^{-1/2}). \quad (6.6)$$

The limit is not a single continuous Gaussian: subsequences with different phases have different discrete endpoint laws.

6.3 Exact modes and a uniform two-endpoint theorem

An exact ratio gives more than a truncated Stirling approximation in the supercritical regime. From (2.2),

$$\rho_m(r) := \frac{(r+2)(m-r)}{r(m+r+2)}, \quad 0 < r < m. \quad (6.7)$$

It equals $b_m(r+2)/b_m(r)$ when both arguments lie in $(0, m]$, and equals 1 exactly when $m = r(r+2)$.

Theorem 6.3 (Exact modes and all supercritical powers). *For $m \geq 3$, set $y = \sqrt{m+1}$, and let r_- be the largest positive integer of parity m satisfying $r_- \leq y$. Put $r_+ = r_- + 2$ and $W_\pm = b_m(r_\pm)$.*

The maximum ballot entry occurs at the admissible r nearest to y . There are exactly two tied maxima if and only if $m+1 = s^2$ for an integer $s \geq 2$; then their locations are $r = s-1, s+1$. For every real $k > 0$,

$$0 \leq \frac{S_{n,k}}{W_-^k + W_+^k} - 1 \leq \sum_{\ell \geq 1} \exp\left(-\frac{2k\ell^2}{m+1}\right). \quad (6.8)$$

Consequently, whenever $k/m \rightarrow \infty$, with no upper bound on k ,

$$S_{n,k} = (W_-^k + W_+^k) \left[1 + O\left(e^{-2k/(m+1)}\right) \right], \quad (6.9)$$

uniformly in the phase.

Proof. The sign of $\rho_m(r) - 1$ is the sign of $m - r(r+2)$, so adjacent entries increase up to the midpoint criterion $r+1 = y$ and decrease after it. This proves the nearest-point and exact-tie statements. The candidate pair lies within $0 < r \leq m$ for $m \geq 3$.

For the uniform remainder, differentiate the logarithm of the exact ratio:

$$\begin{aligned} \frac{d}{dr} \log \rho_m(r) &= -\frac{2}{r(r+2)} - \frac{1}{m-r} - \frac{1}{m+r+2} \\ &\leq -\frac{2}{m+1}, \quad 0 < r < m. \end{aligned}$$

The last inequality follows because $(m - r)(m + r + 2) = (m + 1)^2 - (r + 1)^2 \leq (m + 1)^2$. Also $\log \rho_m(y - 1) = 0$.

For the i th ratio to the left of r_- , the distance from its argument $r_- - 2i$ to $y - 1$ is at least $2i - 1$. Integrating the derivative bound and summing the ratios therefore gives

$$\frac{b_m(r_- - 2\ell)}{W_-} \leq \exp\left(-\frac{2\ell^2}{m + 1}\right)$$

whenever that lattice point is admissible. The ratios to the right of r_+ obey the same estimate relative to W_+ . Raising to k and summing both tails proves (6.8). Finally, with $t = k/(m + 1)$,

$$\sum_{\ell \geq 1} e^{-2t\ell^2} \leq \frac{e^{-2t}}{1 - e^{-6t}},$$

which proves the claimed supercritical order as $t \rightarrow \infty$. $\square$

This theorem keeps the two dominant ballot entries *exact*. It is therefore valid even when errors from raising a Stirling approximation to an extremely large power would be unacceptable. It also identifies a small but consequential distinction: the leading continuous saddle is $\sqrt{m}$, whereas the exact discrete mode is selected by $\sqrt{m + 1}$. A shift negligible at $k \asymp m$ need not be negligible at a much larger exponent.

The two main terms can equivalently be written

$$W_-^k + W_+^k = 2(W_- W_+)^{k/2} \cosh\left(\frac{k}{2} \log \frac{W_+}{W_-}\right). \quad (6.10)$$

Thus the precise crossover is governed by $k \log(W_+/W_-)$, not by an uncontrolled choice of one approximate maximum.

6.4 A second transition near square-index ties

Corollary 6.4 (Near-square logistic limit). *Fix an integer d and $\kappa > 0$. Along*

$$n = s^2 - 2 + 2d, \quad k/s^3 \rightarrow \kappa, \quad s \rightarrow \infty, \quad (6.11)$$

the endpoint law concentrates on $h = s - 2$ and $h = s$, and

$$\mathbb{P}_{n,k}\{h = s\} \rightarrow \frac{1}{1 + e^{-4\kappa d}}, \quad \mathbb{P}_{n,k}\{h = s - 2\} \rightarrow \frac{1}{1 + e^{4\kappa d}}. \quad (6.12)$$

For $k = \kappa s^3 + O(1)$ the errors are $O(s^{-2})$.

Proof. For fixed d and large s , the bracketing admissible points are $r_- = s - 1$ and $r_+ = s + 1$. The factor $2d$ in (6.11) preserves their parity; replacing it by an arbitrary odd shift would change the endpoint lattice. Substitution into (6.7) gives the exact identity

$$\frac{W_+}{W_-} = 1 + \frac{4d}{(s - 1)(s^2 + s + 2d)}, \quad \log \frac{W_+}{W_-} = \frac{4d}{s^3} + O_d(s^{-5}). \quad (6.13)$$

Since $k/m \sim \kappa s \rightarrow \infty$, Theorem 6.3 makes the other endpoints exponentially negligible. The upper endpoint probability is therefore $\rho^k/(1 + \rho^k)$ plus an exponentially small error. Equation (6.13) yields (6.12) and its error bound. $\square$

The first transition, at $k \asymp n$, changes a broad Gaussian sum into a finite-width lattice sum. The second, at $k \asymp n^{3/2}$ near square-index ties, resolves two neighboring entries whose logarithms differ by only order $n^{-3/2}$. At $d = 0$ the exact tie persists and the two limiting probabilities are $1/2$. This second transition is not an assumption about the phase: it follows from the exact ballot ratio.

7 Multisets and exponentially small sectors

7.1 An exact bridge between the two sequences

Let $E_q(n) = e_q(0, 1, \dots, n-1)$ be an elementary symmetric polynomial, so that

$$\prod_{j=0}^{n-1} (X + j) = \sum_{q=0}^n E_q(n) X^{n-q}. \quad (7.1)$$

In particular $E_0 = 1$, $E_1 = \binom{n}{2}$, and $E_n = 0$ for $n \geq 1$. Multiplying each multiset binomial by $n!$ and summing yields the exact identity

$$n!b_n = \sum_{q=0}^n E_q(n) S_{n,n-q}. \quad (7.2)$$

This identity separates the dominant tuple count from its successive corrections without approximating individual small ballot entries.

Theorem 7.1 (Every fixed exponential sector). *For each fixed $Q \geq 0$, as $n \rightarrow \infty$,*

$$\frac{n!b_n}{a_n} = \sum_{q=0}^Q E_q(n) \frac{S_{n,n-q}}{a_n} + O_Q\left(n^{2Q+2} V_m^{-Q-1}\right). \quad (7.3)$$

For each fixed q , the quotient $S_{n,n-q}/a_n$ has a phase-uniform expansion to every algebraic order, with leading term V_m^{-q} . Thus (7.3) is an expansion in successive exponentially small scales, each with its own algebraic expansion.

Proof. Nonnegativity and the multinomial expansion give

$$0 \leq E_q(n) \leq \frac{E_1(n)^q}{q!}.$$

For $0 \leq q \leq n/2$, Theorems 2.2 and 6.2, uniformly for the exponent ratio in a fixed interval around $[1/2, 1]$, imply

$$\frac{S_{n,n-q}}{a_n} \leq C V_m^{-q}.$$

For the remaining indices, let M be the largest ballot entry. The nearest-point estimate and the local expansion give $M/V_m = \exp(O(1/m))$. Since there are at most $n+1$ summands and $a_n \geq M^n$,

$$\frac{S_{n,n-q}}{a_n} \leq (n+1)M^{-q} \leq Cn V_m^{-q}.$$

Put $x = E_1/V_m$, which tends to zero exponentially. Summing the first bound over $q \geq Q+1$ through $n/2$ gives $O_Q(x^{Q+1})$. The remaining tail is at most $Cn \sum_{q>n/2} x^q/q!$, smaller than every fixed power of x for large n . The exact identity (7.2) proves the remainder.

For fixed q , apply the all-orders form of Theorem 6.2 with $\tau = (n - q)/m = 1 - (q + 1)/m$ and Taylor-expand its smooth coefficient functions in τ . Divide by the positive all-orders denominator from Theorem 2.2. Formal series division is justified at each finite order because the leading theta factor is bounded below. The leading factors cancel except for V_m^{-q} . $\square$

7.2 The first sector and its phase correction

Theorem 7.2 (Explicit leading multiset correction). *Let*

$$\Lambda(\alpha) = \frac{M_2(\alpha)}{\Theta(\alpha)} - \frac{1}{6}. \quad (7.4)$$

Then

$$\frac{n!b_n}{a_n} = 1 + \frac{\binom{n}{2}}{V_m} \left[1 + \frac{\Lambda(\alpha_n)}{m} + O(m^{-3/2}) \right] + O(n^4/V_m^2). \quad (7.5)$$

In particular,

$$\boxed{\frac{n!b_n}{a_n} = 1 + \frac{\sqrt{\pi e}}{4\sqrt{2}} n^3 2^{-n} (1 + O(n^{-1}))}. \quad (7.6)$$

Proof. Under the probability measure proportional to $B(n, h)^n$, the ratio $S_{n,n-1}/a_n$ is the expectation of $B(n, h)^{-1}$. The local logarithmic expansion gives

$$\log \frac{b_m(\sqrt{m} + u)}{V_m} = \frac{1/6 - u^2}{m} + O\left(m^{-3/2}(1 + |u|)^3\right)$$

in the summable expansion sense proved in Section 3. Consequently

$$\frac{S_{n,n-1}}{a_n} = V_m^{-1} \left[1 + \frac{M_2(\alpha_n)/\Theta(\alpha_n) - 1/6}{m} + O(m^{-3/2}) \right].$$

There is no $m^{-1/2}$ term in this quotient: the reciprocal ballot entry differs from V_m^{-1} first at order m^{-1} , and the $m^{-1/2}$ change in its weighting affects the expectation only at order $m^{-3/2}$. Polynomial-weighted Gaussian tail control justifies this multiplication and expectation.

Take $Q = 1$ in Theorem 7.1. Finally,

$$\frac{\binom{n}{2}}{V_m} = \frac{\sqrt{\pi e}}{4\sqrt{2}} (n^3 - n) 2^{-n}.$$

The boundedness of Λ and the smaller next sector yield (7.6). $\square$

This proves that $b_n \sim a_n/n!$ with an exponentially small relative error, not merely with an unspecified $o(1)$ error. It also prevents a misleading conclusion: although the two sequences share every algebraic asymptotic order after multiplication by $n!$, they are not identical beyond all orders.

7.3 The OEIS normalization for the multiset sequence

The ordinary reciprocal Stirling expansion gives

$$\frac{\mathcal{A}_n}{n!} = \frac{\mathcal{B}_n}{\sqrt{2\pi}} \left[1 - \frac{1}{12n} + O(n^{-2}) \right]. \quad (7.7)$$

Combining this with the diagonal expansion and the exponential estimate shows

$$\frac{b_n}{\mathcal{B}_n} = \frac{e^{-5/6}}{\sqrt{2\pi}} \left[\Theta(\alpha_n) + m^{-1/2} F_1(\alpha_n) + m^{-1} \left(F_2(\alpha_n) - \frac{\Theta(\alpha_n)}{12} \right) + O(m^{-3/2}) \right]. \quad (7.8)$$

An all-orders version follows by multiplication of the two finite asymptotic expansions. Its exact limiting interval is $[C_-/\sqrt{2\pi}, C_+/\sqrt{2\pi}]$, and its normalized limiting law is that in Corollary 4.4 scaled by $1/\sqrt{2\pi}$. This proves and refines the nonconvergence recorded for A357871. It also gives

$$b_n^{1/n} \sim \frac{e^{1/2} 2^{n+3/2}}{\sqrt{\pi} n^2},$$

agreeing with the existing root formula [2].

Finally, the Turán quotient for b_n differs from that of a_n by $n/(n+1)$ times an exponentially small relative perturbation. Its liminf and limsup are therefore also T_- and T_+ . In particular b_n is eventually strictly log-convex as well.

8 Inverting the oscillatory expansion

8.1 Why an inverse must specify a parity branch

An integer sequence does not come with a canonical continuous inverse. Moreover α_n contains a parity switch, so interpolating its two parity classes by a single differentiable phase is inappropriate. We instead define explicit asymptotic models and quantify how accurately their inverses recover the original integer data.

For $\sigma \in \{0, 1\}$ and real $x > 0$, set

$$\alpha_\sigma(x) = \frac{\sqrt{x+1} - \sigma}{2}. \quad (8.1)$$

The branch appropriate to an integer n has $\sigma = (n+1) \bmod 2$. For each fixed $J \geq 0$, define

$$A_{\sigma,J}(x) = \exp(g(x)) e^{-5/6} \sum_{j=0}^J (x+1)^{-j/2} F_j(\alpha_\sigma(x)). \quad (8.2)$$

This is a definition of a smooth approximating function, not a claim that the exact counting sequence has that interpolation.

Proposition 8.1 (Matched inverse accuracy). *For each fixed J and σ , $A_{\sigma,J}$ is positive and strictly increasing for all sufficiently large x . If $r_{\sigma,J}(L)$ is the large solution of $\log A_{\sigma,J}(x) = L$, then, for integers n of the matching parity,*

$$r_{\sigma,J}(\log a_n) - n = O_J(n^{-(J+3)/2}). \quad (8.3)$$

Proof. Positivity follows from the positive lower bound on Θ and the boundedness of all coefficient functions. Differentiating (8.2), and using $\alpha'_\sigma(x) = 1/(4\sqrt{x+1})$, gives

$$\frac{d}{dx} \log A_{\sigma,J}(x) = 2\lambda x - \log x - 1 + c + O(x^{-1/2}) = 2\lambda x + O(\log x) > 0.$$

Theorem 2.2 gives a logarithmic residual at $x = n$ of order $n^{-(J+1)/2}$. The mean value theorem divides that residual by a quantity comparable to n , proving (8.3). $\square$

Define the first logarithmic coefficient functions

$$d_0(\alpha) = -\frac{5}{6} + \log \Theta(\alpha), \quad d_1(\alpha) = \frac{F_1(\alpha)}{\Theta(\alpha)}. \quad (8.4)$$

Thus

$$\log a_n = g(n) + d_0(\alpha_n) + m^{-1/2}d_1(\alpha_n) + O(m^{-1}).$$

Higher logarithmic coefficients are obtained by ordinary finite series expansion of the logarithm; for example $d_2 = F_2/\Theta - F_1^2/(2\Theta^2)$.

8.2 An explicit inverse through the first oscillatory orders

Theorem 8.2 (Parity-sensitive inverse expansion). *Let $L \rightarrow \infty$, and write*

$$s = \sqrt{\frac{L}{\lambda}}, \quad d = \frac{\log s - c}{2\lambda}, \quad \beta = \alpha_\sigma(s + d). \quad (8.5)$$

For any fixed $J \geq 1$,

$$r_{\sigma,J}(L) = s + d + \frac{\lambda d^2 + d - d_0(\beta)}{2\lambda s} - \frac{d_1(\beta)}{2\lambda s^{3/2}} + O\left(\frac{(\log s)^2}{s^2}\right). \quad (8.6)$$

The implied constant is uniform in the two parity branches.

Proof. Since $\lambda s^2 = L$ and $2\lambda d = \log s - c$, Taylor expansion gives

$$g(s + d) - L = -\lambda d^2 - d + O(d^2/s), \quad g'(s + d) = 2\lambda s - 1 + O(d/s).$$

The logarithm of the model at $s + d$ adds $d_0(\beta) + s^{-1/2}d_1(\beta) + O((1 + d)/s)$. A Newton correction using the leading derivative $2\lambda s$ is exactly the two displayed correction terms in (8.6). Its remaining logarithmic residual is $O((1 + d)^2/s)$, including the derivative correction and the omitted model terms. Phase drift during this correction is bounded by the correction times $O(s^{-1/2})$ and is smaller than this residual. Dividing once more by a derivative comparable to s proves the stated error, because $d = O(\log s)$. $\square$

The phase in this formula is evaluated at $s + d$, not merely at s . That choice incorporates the logarithmic shift before applying the oscillatory correction. Dropping $d_0(\beta)$ discards an order- s^{-1} term; replacing it by its average cannot recover individual inverse values at that order.

For any requested finite accuracy, compute more F_j , form the logarithm of (8.2), and apply Newton iteration or formal coefficient reversion. At large x , the derivative is of order x and the second derivative is bounded. Hence a Newton step with an input error $e = o(x)$ has output error $O(e^2/x)$ relative to the chosen model root. Along with Proposition 8.1, this gives an effective all-orders inversion procedure. It does not assert convergence of an infinite formal inverse expansion.

8.3 Integer thresholds and the multiset inverse

For $y > 1$, let

$$N(y) = \min\{n \geq 1 : a_n \geq y\}.$$

The sequence is strictly increasing from $n = 1$ onward. One direct injection extends every path in an ordered n -tuple by an up-step and adds a duplicate of the first extended path as the last component. The original tuple is recovered from the first n components. For $n \geq 1$, a tuple of $(n + 1)$ -step paths ending in a down-step exists and is outside the image, proving strictness. Also $a_0 = a_1 = 1$.

The two model roots provide candidate thresholds on their respective parity lattices. Rounding a model root upward to the matching parity is correct whenever its distance from the relevant integer boundary exceeds the root error bound. Near a boundary, an exact comparison is needed. Thus (8.3) is a subinteger accuracy statement for matched continuous models, *not* an unconditional exact rounding theorem for all real y .

For the multiset sequence the large logarithmic scale becomes

$$g_b(x) = \lambda x^2 - 2x \log x + (c + 1)x - \frac{1}{2} \log x - \frac{1}{2} \log(2\pi). \quad (8.7)$$

One combines it with the same d_0, d_1 , the reciprocal-Stirling corrections, and, when required, the logarithm of the exponential-sector factor in (7.5). Its derivative is again $2\lambda x + O(\log x)$, so the same branchwise inversion and error argument applies. An exponential relative correction of order $x^3 2^{-x}$ changes the inverse by order $x^2 2^{-x}$, smaller than every algebraic inverse term.

9 Computational verification and reproducibility

9.1 What was checked exactly

The companion program `code/verify.py` separates exact integer assertions, symbolic algebra, and high-precision numerical calculations. All reported checks completed successfully. An independent dynamic program for nonnegative paths was compared with the reflection formula and the shifted binomial formula through $n = 100$: this tests 2601 ballot entries. The first 15 entries of each target sequence were compared with the OEIS data [1, 2, 5, 6]. The rising-factorial identity was tested exactly through $n = 25$; the fixed-power identities for $k = 1, 2$ through $n = 100$; and the adjacent-entry ratio through $n = 100$. Exact tied maxima were checked for every $n = s^2 - 2$ with $2 \leq s \leq 20$.

The coefficient generator uses rational symbolic arithmetic for (3.3), (3.7), and (3.10). It produces $R_0, \dots, R_6$, verifies their parities, and checks the first displayed polynomial formulas. These finite checks can catch an index shift or transcription error, but do not establish a uniform asymptotic theorem. The Gaussian bounds in Section 3 are the mathematical justification of that theorem.

9.2 Accuracy of the all-orders approximation

For selected indices, the numerical reference is the full sum over *all* endpoints, using exact integer ballot entries as bases and 75-decimal-digit logarithmic evaluation. The approximate value is the truncated expansion in Theorem 2.2. The signed error below is approximation divided by numerical reference, minus one. These are high-precision experiments, not interval-arithmetic certificates.

At $n = 1200$ the order-6 approximation has relative error about $7.52 \cdot 10^{-12}$. At square-adjacent indices some odd coefficient functions are small, and errors can be unusually favorable. No monotone improvement in J for every individual n is asserted. An asymptotic expansion controls fixed orders as n grows, not arbitrary truncation at a fixed small index.

n	$J = 0$	$J = 2$	$J = 4$	$J = 6$
25	$-5.892 \cdot 10^{-2}$	$-1.503 \cdot 10^{-4}$	$2.293 \cdot 10^{-5}$	$2.462 \cdot 10^{-6}$
50	$-6.268 \cdot 10^{-3}$	$-5.377 \cdot 10^{-4}$	$-1.202 \cdot 10^{-5}$	$-3.373 \cdot 10^{-7}$
100	$-1.608 \cdot 10^{-2}$	$-1.198 \cdot 10^{-5}$	$3.781 \cdot 10^{-7}$	$1.145 \cdot 10^{-8}$
200	$-1.075 \cdot 10^{-2}$	$6.415 \cdot 10^{-5}$	$4.041 \cdot 10^{-7}$	$2.464 \cdot 10^{-9}$
400	$-4.115 \cdot 10^{-3}$	$-7.936 \cdot 10^{-7}$	$5.982 \cdot 10^{-9}$	$4.673 \cdot 10^{-11}$
800	$-4.439 \cdot 10^{-3}$	$1.415 \cdot 10^{-5}$	$2.023 \cdot 10^{-8}$	$2.850 \cdot 10^{-11}$
1200	$-2.722 \cdot 10^{-3}$	$6.828 \cdot 10^{-6}$	$6.878 \cdot 10^{-9}$	$7.520 \cdot 10^{-12}$

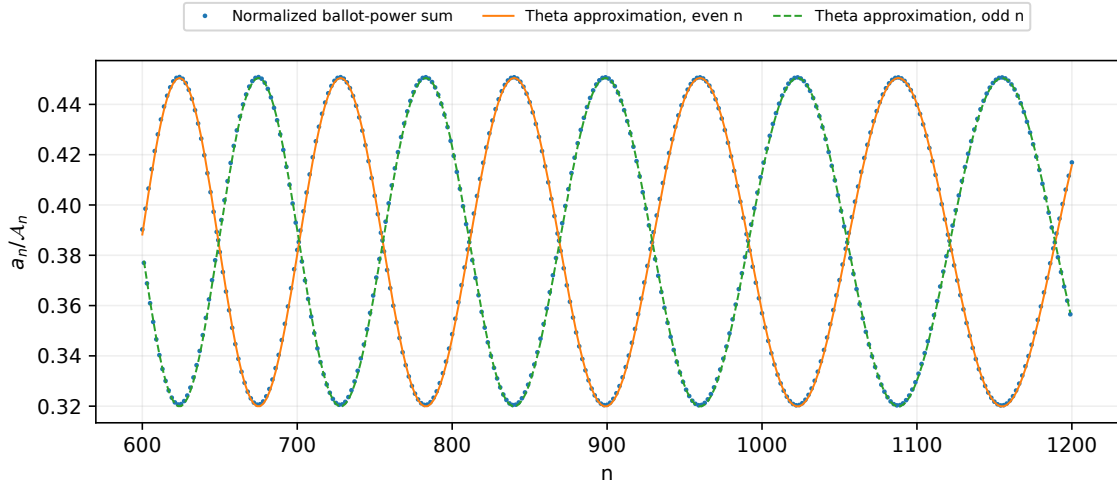
Table 1: Signed relative errors for truncations through order $m^{-J/2}$.

Figure 2: The normalized diagonal sequence and its leading approximation. Connecting the even and odd indices separately reveals the two smooth parity strands. The index-dependent widening of an oscillation is consistent with the square-root phase. Plotting computations use a localized log-Gamma sum; the full-sum tests are reported separately.

9.3 Exponential corrections, near-square transitions, and inverses

The first exponential sector was checked using exact integers for a_n , b_n , and $n!$. For $n = 20, 30, 40, 60, 80, 100$, the ratio of the exact relative correction to $\sqrt{\pi e} n^{3/2} 2^{-n} / (4\sqrt{2})$ ranges from approximately 0.99969 to 1.01631. Its order- $1/n$ phase dependence, rather than a single monotone convergence curve, is predicted by (7.5). The complete values are in `data/exponential_sector.csv`.

For the second localization transition, the code evaluates full sums at $s = 8, 16, 32$, $d = -2, -1, 0, 1, 2$, and $k = \text{round}(0.3s^3)$. The resulting upper-endpoint probabilities are compared with $(1 + \exp(-1.2d))^{-1}$ in `data/square_transition.csv`. The precise choice of the parity-preserving window $n = s^2 - 2 + 2d$ is tested rather than replaced by a nearby generic index.

The inverse computations use $L = \log a_n$ from the full endpoint sum. They compare (8.6) with a numerical root of the order-6 model of matching parity. Representative signed errors are:

n	Explicit inverse minus n	Order-6 model root minus n
100	$-9.714 \cdot 10^{-4}$	$-8.609 \cdot 10^{-11}$
200	$-3.112 \cdot 10^{-4}$	$-9.096 \cdot 10^{-12}$
400	$-9.734 \cdot 10^{-5}$	$-8.535 \cdot 10^{-14}$
800	$-2.994 \cdot 10^{-5}$	$-2.588 \cdot 10^{-14}$
1200	$-1.506 \cdot 10^{-5}$	$-4.543 \cdot 10^{-15}$

These measurements illustrate the extra factor of order $1/n$ gained when a logarithmic approximation is inverted. They do not provide a certified least threshold for the big- O statements.

9.4 Reproducing the files

From the extracted archive, run

```
python -m pip install -r requirements.txt
python code/verify.py --all
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The code uses Python 3.10 or newer with SymPy, mpmath, and matplotlib. The verification itself does not require network access; the OEIS values used in exact comparisons are embedded with their provenance. The first command is only for installing missing dependencies. Alternatively, `build.sh` compiles the article when the dependencies are already installed. The archive includes the figures used by the TeX source, raw CSV tables, generated coefficient polynomials, and the actual verification-status JSON.

No Lean file is included, because no formal proof was compiled for this report. No interval certificate is implied by a large working precision. The existing repo was inspected but not modified.

[Write note, batch 73O1.] In the collection the requirements file is `data/requirements.txt` and the build script is `code/build.sh`. The script changes into its own directory before calling `code/verify.py`, so it runs only in the delivered layout. `code/verify.py --all` rewrites `data/` and `figures/` in place, so the shipped records should be regenerated on a copy; the README gives commands that do this.

10 Further research questions and concrete next steps

The following questions are distinguished from the theorems already proved. They suggest a program extending the results rather than merely repeating the missing-constant question.

- Q1. Effective remainders and certified OEIS additions.** Make the constants in (3.11) explicit at orders $J = 0, 2, 4$, and combine them with interval evaluation of the theta sums. This would yield rigorous numerical enclosures for every sufficiently large index, an explicit log-convexity threshold, and certified threshold inversion. A finite exact check could then bridge the remaining initial range. The global trigamma estimate and the explicit two-site bound already provide useful non-asymptotic ingredients.
- Q2. Global strict log-convexity.** Do $a_{n-1}a_{n+1} > a_n^2$ and $b_{n-1}b_{n+1} > b_n^2$ hold at every $n \geq 1$? The limiting Turán gap proves eventual strictness, but does not settle the finite range

without explicit error constants. A direct injection, total-positivity argument, or inequality linking adjacent ballot rows might prove the global statement without enumeration.

- Q3. Uniform secondary transition expansions.** The exact two-site formula resolves the entire supercritical regime, and Corollary 6.4 resolves bounded square-window shifts. Develop uniform expansions when the integer displacement d grows with s , or when k/s^3 tends to zero or infinity simultaneously with d . The controlling variable is the exact $k \log \rho$, so a meaningful goal is a hierarchy of elementary scaled formulas with explicit error domains, not replacing an exact expression by an uncontrolled Taylor truncation.
- Q4. Optimal truncation and additional exponential scales.** The all-orders diagonal expansion is not an optimal-truncation theorem. Determine the large-order growth of the polynomials and theta-derivative coefficients, and whether a rigorously optimally truncated remainder can be decomposed into further exponential contributions. This problem is separate from the rigorously established tuple-multiset sectors; it should not be called resurgence without additional analysis.
- Q5. A growing number of multiset sectors.** Theorem 7.1 keeps Q fixed. Find useful ranges of $Q = Q(n)$ with uniform coefficient and remainder control, and study multisets whose size differs from the path length. The elementary symmetric coefficients of $0, \dots, n-1$ and the ratio k/m then interact in a two-parameter problem. An exact rising-factorial bridge remains available, but uniform asymptotic division requires new estimates.
- Q6. A reusable lattice-peak theorem for other path models.** Formulate sufficient hypotheses on a positive triangular array $C_{n,h}$ for a high-power sum to have a phase-dependent theta expansion. The proof here suggests the required data: a uniformly concave local logarithm, a growing interior peak, a fixed endpoint lattice, derivative expansions, and global tail domination. Apply such a theorem to weighted meanders and finite-step walks, then to multidimensional endpoint lattices where theta functions are associated with quadratic forms. This is the clearest route from the present OEIS case to a broad tool.
- Q7. Joint phase laws and higher discrete differences.** Determine effective joint laws of several neighboring normalized terms, and use them to analyze higher logarithmic differences or Jensen polynomials. Adjacent terms undergo a half-period shift rather than an independent random phase. The second difference is positive because its explicit theta quotient has a gap; analogous sign or hyperbolicity claims at higher orders require separate proofs.
- Q8. Sharper averaging errors.** The elementary discrepancy proof gives $O(\sqrt{N})$ counting error. For smooth periodic test functions, exploit the square-root phase, parity cancellation, and vanishing odd corrections to seek sharper Cesàro expansions. Such expansions may contain boundary-phase terms rather than a universal constant at the next order. An unsmoothed interval discrepancy and a smooth weighted average need not have the same best error term.
- Q9. The separate arithmetic conjecture.** The OEIS records Bala's conjecture $a_{2p-1} \equiv 1 \pmod{p^3}$ for primes $p \geq 5$ [1]. Nothing in the present asymptotic proof establishes it. The exact shifted ballot formula gives a possible starting point for a prime-adic analysis, but real asymptotic errors, even exponentially small relative errors, have no direct force modulo p^3 . A future arithmetic report should maintain that distinction explicitly.
- Q10. Formalization with a visible theorem boundary.** Start with exact path semantics, reflection, shifted binomial identities, the adjacent-entry ratio, and the rising-factorial

bridge. Next formalize the discrete two-site tail bound, which avoids Stirling theory. Only then build the analytic layer: gamma remainders, summable Gaussian bounds, phase-uniform termwise expansion, and theta identities. This staged plan could produce useful Lean theorems before the entire analytic result is available, without labeling the unformalized remainder as checked.

11 Conclusion

The central obstacle was not the exponential scale of the counts. That scale was already visible in the OEIS. The missing information lay in the interaction between a moving ballot peak and an endpoint lattice that remains coarse after taking a power proportional to the row index. Keeping that lattice produces an explicit theta multiplier, not a constant, and supports a complete algebraic expansion with uniform remainders.

The same analysis produces exact fluctuation envelopes and phase laws, strict eventual log-convexity, and stable branchwise inverse formulas. An exact adjacent-entry ratio further identifies the mode and controls all supercritical powers, including a second near-square transition. Finally, an exact symmetric-polynomial identity places the multiset sequence into a hierarchy of exponentially small corrections to the ordered-tuple sequence. Together these results turn a recorded nonconvergence phenomenon into a coherent analytic and combinatorial structure, while leaving the distinct arithmetic and formalization questions clearly identified.

A Additional explicit coefficients

For direct implementation, the next two multiplier polynomials are

$$R_3(u) = \frac{u^9}{162} - \frac{5u^7}{108} + \frac{119u^5}{270} + \frac{197u^3}{1620} - \frac{7u}{18}, \quad (\text{A.1})$$

$$R_4(u) = \frac{u^{12}}{1944} - \frac{19u^{10}}{1944} + \frac{1429u^8}{12960} - \frac{3017u^6}{9720} + \frac{5227u^4}{19440} + \frac{311u^2}{540} + \frac{8263}{50400}. \quad (\text{A.2})$$

The next two Stirling coefficient polynomials are

$$A_4(x) = -\frac{x^{10}}{90} + \frac{x^8}{8} - \frac{x^6}{3} + \frac{4x^2}{15}, \quad (\text{A.3})$$

$$A_5(x) = -\frac{x^{12}}{132} + \frac{x^{10}}{10} - \frac{x^8}{3} + \frac{2x^4}{3} - \frac{1}{20}. \quad (\text{A.4})$$

The archive supplies D_j and R_j through $j = 6$ in `data/coefficients.json`. The source function `symbolic(order)` can generate higher specified orders; its numerical evaluator is intentionally configured for the supplied orders 0 through 6 and rejects unsupported numerical truncations.

The following identities can be used instead of differentiating theta series numerically:

$$M_1 = \frac{\Theta'}{4}, \quad M_2 = \frac{\Theta''}{16} + \frac{\Theta}{2}, \quad M_3 = \frac{\Theta'''}{64} + \frac{3\Theta'}{8}.$$

For τ near 1, direct Gaussian sums are especially efficient. For small τ , the Fourier representation (4.1) is preferable because its nonzero modes are exponentially suppressed. Whichever

representation is used, rigorously certified computation requires an explicit tail bound and rounding control; the numerical script uses high precision but does not claim interval certification.

B A compact dependency and trust map

The logical dependency chain is:

$$\text{ballot identity} \longrightarrow \begin{cases} \text{Stirling expansion + concavity} \longrightarrow \text{theta expansion,} \\ \text{exact entry ratio} \longrightarrow \text{two-site theorem,} \\ \text{rising factorial} \longrightarrow \text{multiset sectors.} \end{cases}$$

The theta expansion plus elementary square-root phase counting proves the limiting interval, distribution, and quotient statements. Smooth finite model inversion proves the inverse results. The multiset-sector proof also uses the uniform critical estimate. No finite numerical test is an assumption in these implications.

The exact computations test definitions and coefficient algebra; the numerical tables test consequences. Neither is a premise of an analytic theorem. The standard gamma and theta inputs are cited explicitly. No conjecture from ProveIt or OEIS is imported as a lemma.

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